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PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	The Columnar to Equiaxed Transition and Dendrite Fragmentation		
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PI Email:	p-voorhees@northwestern.edu	**Fax:**	FY 847-491-7820
PI Organization Type:	UNIVERSITY	**Phone:**	847-491-7815
Organization Name:	Northwestern University		
PI Address 1:	Department for Materials Science and Engineering		
PI Address 2:	2220 Campus Dr		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
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Contact Monitor:	Rabenberg, Ellen	**Contact Phone:**	(256) 544-7264
Contact Email:	ellen.m.rabenberg@nasa.gov		
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Task Description:	The shape and size of dendritic grain structures in engineering alloys is central to the resulting properties of the material. For example, equiaxed grain structures tend to provide more isotropic materials properties then materials with columnar grain morphologies. By contrast, in other applications, such as NI-base superalloys, columnar grain structures provide superior high temperature or creep properties. It is thus of critical importance to understand and predict the factors affecting the formation of grain morphology during solidification. The processes controlling the formation of equiaxed and columnar grains during solidification will be investigated using two approaches. The first is to measure the grain structures of Al-Si alloys processed by the Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) team that is sponsored by the European Space Agency (ESA), and the second is to examine the effects of solidification on the dendrite fragmentation process. By processing these samples on the International Space Station (ISS), both sedimentation of the dendritic fragments and buoyancy driven convection can be avoided. This will permit an understanding of the physics underlying these processes to be gained, and enable a careful comparison with simulation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In nearly all casting processes, the resulting solid consists of an array of grains of different crystallographic orientation. Depending on the processing parameters, columnar or equiaxed grain growth is present, while in others there is a transition from columnar to equiaxed grains (CET). The importance of the CET lies in the clear connection between the properties of the solidified ingots and the grain structure. For example, ingots with elongated grains can have much different creep behavior than those with equiaxed grains. The grains in this case can be a single dendrite, as in the equiaxed region, or a collection of dendrites with the same crystallographic orientation. In many cases, such as in hypoeutectic alloys during cooling a mush of dendrites and liquid is formed. The last to solidify liquid forms a eutectic both between the secondary dendrite arms, but more importantly in equiaxed structures between the grains. This can lead to casting defects, such as hot tearing and intergranular cracking. Thus, the conditions under which an ingot solidifies with a columnar or equiaxed structure can have a major effect on the properties of the resulting part. Moreover, if the conditions under which the CET occurs are well understood, a properly placed CET can be used to tailor the properties of the ingot, as well as the spatial distribution of the intergranular eutectic.
Task Progress:	The science requirements document (SRD) for Dendrite Fragmentation and Morphology During Melting and Solidification (DFM) has been approved and the samples for DFM have been prepared and are currently on board the International Space Station awaiting processing. In addition, samples from the Columnar-to-Equiaxed Transition in Solidification processing (CETSOL) microgravity experiments have been analyzed. A. DFM The DFM SRD was completed and authorized, detailing the data we hope to obtain from the experiments. The experiments will coarsen samples during set heating and cooling rates. The purpose of looking at coarsening with different heating or cooling rates follows the work outlined by Neumann-Heyme and coworkers (2015) who predict that there exist critical heating/cooling rates separating the regimes of dendrite fragmentation, coalescence, and retraction. The dominant behavior could have an effect on the number of fragments available to serve as nucleation sites for equiaxed grains, affecting the CET and the microstructure of the solidified part at large. The DFM project will probe different heating and cooling rates to determine the existence or value of these critical values. Calculations were performed and are presented in the SRD, showing that that the microgravity environment, the dendrite size and shape, the sample size and shape, and the temperature profiles selected for DFM are expected to yield the desired data to investigate this process. Additional calculations in the SRD also show, based on the temperature profiles and samples chosen, these samples should allow for reasonable comparisons to previous samples from the Coarsening in Solid-Liquid Mixtures (CSLM) project. The samples to be used for the DFM project have also been created. Rods of material containing the appropriate volume fractions of dendrites (40, 50, and 60 vol%) were created and then melted and re-solidified using a Bridgman-type furnace with an applied axial rotation to reduce sedimentation of the dendrites. The solidification velocity and gradient chosen produced dendrites with a similar shape and size scale to previous CSLM experiments. Once these rods were re-solidified, we noticed that the volume fraction of a sample taken from a rod varied depending on where along its length it was located. To ensure that the samples had the desired volume fraction and were homogeneous throughout their axial length, optical images were captured and segmented for both sides of every sample that was to be used as a space flight sample, ground sample, or backup sample. This will also allow us to compare size scale and morphology in 2D between the original samples and the final samples after experimentation. These samples were then marked to indicate which side was which and were transported in a cooler to the engineering team who assembled the ampoules. The flight ampoules are now on board the International Space Station. B. CETSOL Samples from the CETSOL experiment have been analyzed to determine how the different processing parameters and addition of grain refiner affects the formation of fragments and a CET. Sample image data were aligned, segmented, and filtered and processed to obtain data on fragments that are as accurate as possible. Preliminary findings indicate that more fragments have been observed when the primary arms are more misaligned with the solidification direction. An additional sample was recently sectioned and is being segmented and analyzed to verify some of these initial findings. Reference: Neumann-Heyme N, Eckert K, and Beckermann C. 2015. “Dendrite fragmentation in alloy solidification due to sidearm pinch-off,” Phys. Rev. E - Stat. Nonlinear, Soft Matter Phys., vol. 92, no. 6.
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Stan T, Thompson ZT, Voorhees PW. "Building towards a universal neural network to segment large materials science imaging datasets." Proc SPIE. 2019 Sep;11113G. [Developments in X-Ray Tomography XII, (10 September 2019)] https://doi.org/10.1117/12.2525290 , Sep-2019